Exact-Quota Mixed-Precision FIR Quantization: Dynamic Programming, FPGA Validation, and Response-Aware Feasibility

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***Abstract*—This technical report evaluates exact-quota mixed-precision FIR coefficient quantization using 16-, 8-, and 4-bit widths. A dynamic program (DP) minimizes coefficient squared error under exact width counts. The formulation is equivalent to unit-capacity transportation and specializes known cost-constrained quantization methods; no new optimization principle is claimed. Software validation comprises 29 exhaustive checks and 108 configurations compared with greedy and assignment-MILP baselines. Hardware evidence includes six routed 8-tap runs and 942 bit-exact board comparisons, plus four routed 48-tap runs and 14,888 RTL output comparisons. Across five filter designs and 15 design/quota combinations, DP meets both passband and stopband limits in two combinations, while greedy meets them in none. Greedy nevertheless has a lower stopband peak in some failed cases: coefficient-error optimality does not guarantee spectral compliance. A band-weighted response objective introduces cross-terms; its diagonal approximation does not change the allocation. A 48-tap binary-product MILP attempt was manually stopped after about 93 seconds under a pre-declared slow-progress condition, without an optimality certificate or final allocation. The report provides reproducible code and data, including negative findings. The 48-tap study includes no board, power, or Fmax measurements.**

***Index Terms*—FIR filter, coefficient quantization, dynamic programming, mixed-precision, FPGA, assignment problem.**

I. Introduction

Finite-word-length implementation of FIR filters requires assigning a coefficient precision to each tap. When the available hardware supports a discrete set of widths—for example, 16-bit, 8-bit, and 4-bit multipliers—and the system specifies exact counts at each width, the assignment becomes a combinatorial optimization problem. The natural objective in coefficient-domain design is to minimize the total squared coefficient quantization error.

This report studies the exact-quota version of that problem: given N tap coefficients and integer counts n_{16}, n_8, n_4 with $n_{16} + n_8 + n_4 = N$, assign each coefficient a width from $\{16, 8, 4\}$ to minimize

$$\min_{\mathbf{w}} \sum_{i=1}^{N} \big(h_i - Q(h_i, w_i)\big)^2 \tag{1}$$

subject to $|\{i : w_i = w\}| = n_w$ for each width w.

A DP recurrence processes coefficients in order and achieves the global minimum for (1). The formulation is equivalent to a unit-capacity bipartite transportation problem or binary assignment MILP, a relationship established in Section III and verified empirically in Section V.

The report is organized as follows. Section II reviews related work including the closest prior formulations. Section III states the problem and its equivalences. Section IV describes the DP and greedy algorithm. Section V reports software validation. Section VI describes the FPGA case studies. Section VII reports the robustness study. Section VIII documents the response-aware allocation investigation and its negative result. Section IX concludes.

II. Related Work

Several lines of prior work are directly relevant. Twelve papers were reviewed in full in the project literature assessment, and six additional works received abstract-level screening. The reviewed set is targeted, not an exhaustive search of the field.

A. Finite-Word-Length FIR Design

Kodek and Steiglitz [1] analyzed length-wordlength tradeoffs for FIR filters, proving a nonzero lower bound on weighted Chebyshev error for fixed coefficient wordlength. Their complexity proxy Nb motivates joint optimization; their analysis assumes a common wordlength, not per-coefficient quotas.

Lim, Parker, and Constantinides [2] minimized FIR designs over discrete coefficient-value spaces using integer linear and quadratic programming, comparing discrete-space designs with rounded infinite-wordlength designs. Lim [3] extended this to normalized peak-ripple minimization, using MILP methods with passband gain as a design variable.

B. SPT-Based and Hardware-Aware Coefficient Allocation

Several methods allocate signed-power-of-two (SPT) terms across coefficients. Llorens, Hadjicostis, and Ni [4] formulated FIR coefficient quantization under a scalar total integer-cost constraint using a constrained-shortest-path DP. Their alpha=2 case relates coefficient-vector error to frequency-response MSE via Parseval's theorem. For exact three-width quotas, a radix encoding ($r_{16} = B + 1$, $r_8 = 2$, $r_4 = 1$, $B = N + 1$) reduces the quota DP to a scalar exact-cost DP, so the present quota DP is a specialized realization of Llorens et al.'s framework, not a new optimization class. They also describe a greedy algorithm with a conditional optimality guarantee: the guarantee depends on the marginal

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error reduction as the integer cost increases. The sparse costs used in the radix encoding do not automatically satisfy that condition: intermediate cost levels represent no additional width choice, while the final jump to 16 bits can have a positive error reduction. The greedy baseline in this report instead selects the largest immediate squared-error reduction among quota-feasible 4-to-8 and 8-to-16 upgrades. It is a comparator only; no optimality guarantee is claimed, and it is not identified with Llorens et al.'s conditionally optimal greedy algorithm.

Chen and Willson [5] used a trellis (Viterbi-like) search for SPT term allocation against frequency-response criteria; their survivor pruning does not guarantee global optimality. Li, Lim, Lian, and Song [6] allocated a global SPT-term budget with variable per-coefficient counts using greedy sequential allocation; the authors describe the result as suboptimal.

Aktan, Yurdakul, and Dündar [7] used branch-and-bound (FIRGAM) with LP for coefficient bounds to reduce SPT count while meeting frequency specifications; worst-case runtime is exponential.

Kuo, Lin, and Liu [8] integrated successive SPT allocation with common-subexpression elimination under an exact global post-CSE addition budget, reporting 0.35-μm CMOS synthesis results.

C. Subexpression-Space and Structure Optimization

Yu and Lim [9] optimized FIR coefficients directly in a discrete subexpression space using branch-and-bound MILP, minimizing frequency-response ripple subject to an adder budget. This is a direct coefficient-space search, not a two-stage quantize-then-CSE pipeline.

Blad and Gustafsson [10] used ILP to optimize partial-product reduction trees and pipeline registers for fixed coefficient representations, targeting 90-nm CMOS. Their decision variables are hardware structural elements, not coefficient widths.

D. Mixed-Precision FIR Allocation

Fang, Chen, Chen, Wang, and You [11] proposed a PSO-based bit-allocation framework applied to FIR filter design with a frequency-domain minimax objective and aggregate bit-consumption constraint. Their constraint is an inequality on total bits ($\sum b_n \le N\bar{b}$), not exact per-width counts. Their allocation assumes symmetric widths (linear-phase structure). This is the closest prior mixed-precision FIR allocation paper; it differs from the present work in constraint type, objective, and solution method.

Sarma, Sai, and Chatterjee [12] used a coefficient-magnitude threshold to select exact or LUT-approximate multipliers under an output-error budget on a Zynq UltraScale+ radar DDC implementation. Their quantization model (approximate arithmetic) differs from fixed-point coefficient widths.

E. Neural-Network Mixed-Precision

Yang and Jin's FracBits [13] assigns fractional bit-widths in neural-network quantization. It is included as methodological context; it is not a direct FIR baseline.

III. Problem Formulation

A. Exact-Quota Width Assignment

Let $\mathbf{h} = [h_1, \ldots, h_N]$ be a fixed real-valued FIR coefficient vector and let $\mathcal{W} = \{16, 8, 4\}$ be the set of allowed binary widths. Given integer quotas n_{16}, n_8, n_4 with $n_{16}+n_8+n_4 = N$, the problem is to find the width assignment $\mathbf{w} \in \mathcal{W}^N$ minimizing (1) subject to

$$|\{i : w_i = w\}| = n_w, \quad w \in \mathcal{W}. \tag{2}$$

The quantization operator $Q(h, w)$ rounds h to the nearest value representable with w bits (1 sign, $w-1$ fractional bits, no saturation within the tested range).

B. Equivalence to Unit-Capacity Transportation

Problem (1)–(2) is a binary assignment MILP. Let $x_{i,k} \in \{0,1\}$ indicate that coefficient i receives width choices(k); the cost of assignment $x_{i,k} = 1$ is $c_{i,k} = (h_i - Q(h_i, \text{choices}(k)))^2$. Then:

$$\min_{\mathbf{x}} \quad \sum_{i=1}^{N}\sum_{k=1}^{3} c_{i,k}x_{i,k} \tag{3}$$

$$\text{s.t.} \quad \sum_{k} x_{i,k} = 1, \quad i = 1, \ldots, N \tag{4}$$

$$\sum_{i} x_{i,k} = n_k, \quad k = 1, 2, 3 \tag{5}$$

$$x_{i,k} \in \{0, 1\} \tag{6}$$

is a unit-capacity bipartite transportation / min-cost flow problem. The network has a source connected to each of the N coefficient nodes (capacity 1, cost 0), each coefficient node connected to each of the 3 width nodes (capacity 1, cost $c_{i,k}$), and each width node connected to a sink (capacity n_k, cost 0). Sending N units of flow minimizes total cost and satisfies all quotas exactly.

The LP relaxation obtained by replacing $x_{i,k} \in \{0,1\}$ with $x_{i,k} \ge 0$ has integral extreme points: after multiplying one partition of constraint rows by -1, its constraint matrix is a directed bipartite node–arc incidence matrix and is totally unimodular. Since the assignment and quota right-hand sides are integers, an optimal LP extreme point is integral. Thus the assignment formulation can also be solved as a linear program; the reported MILP runs are implementation-level cross-checks, not evidence that integrality requires branch-and-bound.

C. Reduction to Scalar Exact Integer Cost

Llorens et al. [4] study the minimization of an L-alpha coefficient-vector error under a scalar exact integer total cost $\sum_i r_{w_i} = R^*$, solved by a constrained-shortest-path DP. The present three-width exact-quota problem is a special case. Set $B = N + 1$, $r_{16} = B + 1$, $r_8 = 2$, $r_4 = 1$. Then for any assignment, $R = N + Bn_{16} + n_8$; because $0 \le n_8 < B$, the value $R^* = N + Bn_{16} + n_8$ uniquely encodes the quotas (n_{16}, n_8, n_4). The Llorens DP at exact cost R^* therefore solves the quota-constrained problem. This was empirically verified on all 108 saved test configurations (maximum objective discrepancy: floating-point rounding).

The DP developed in Section IV is a direct state encoding that tracks used quota counts rather than scalar cost; it is equivalent in result but distinct in state representation. Neither formulation constitutes a new optimization class.

Scope of the quotas. The counts constrain coefficient representations, not measured LUT, DSP, power, or throughput budgets. The selected 16/8/4-bit quotas are experimental settings; this study does not establish a product-level requirement for those exact counts or a fixed inventory of physical multipliers. Hardware synthesis may remove zero constants or map equal counts differently. Solving the allocation problem therefore does not solve the broader hardware resource–response design problem.

IV. DP Algorithm and Greedy Baseline

A. Dynamic Programming

The DP processes coefficients in order and stores the minimum partial cost for each reached quota state. Let $D[i][c_{16}][c_8][c_4]$ denote the minimum squared error for the first i coefficients using exactly c_{16} 16-bit, c_8 8-bit, and c_4 4-bit assignments. The recurrence is:

$$\begin{aligned} D[i][c_{16}][c_8][c_4] = \min\big\{ & D[i-1][c_{16}-1][c_8][c_4] + e_i(16), \\ & D[i-1][c_{16}][c_8-1][c_4] + e_i(8), \\ & D[i-1][c_{16}][c_8][c_4-1] + e_i(4) \big\} \end{aligned} \tag{7}$$

where $e_i(w) = (h_i - Q(h_i, w))^2$, with the constraint that each dimension of the predecessor state is non-negative.

Initialization is $D[0][0][0][0] = 0$; all other states at $i = 0$ are $+\infty$. The implementation allocates full four-dimensional cost and backpointer arrays, with $(N+1)(n_{16}+1)(n_8+1)(n_4+1)$ entries each. Consequently its implemented space is $O(N(n_{16}+1)(n_8+1)(n_4+1))$. A value-only implementation could use two coefficient layers, but recovering the assignment also requires predecessor information.

The time complexity is $O(N(n_{16}+1)(n_8+1)(n_4+1))$ as an upper bound; each reachable state has at most three transitions. Since $c_4 = i - c_{16} - c_8$, one quota coordinate is redundant, so this product bound is not tight for reachable states.

Optimality. By induction on i: assume $D[i-1]$ stores the optimal cost for each reachable quota state. At step i, the recurrence examines every feasible predecessor; because the objective is separable ($\sum_i e_i(w_i)$) and each coefficient chooses exactly one width, the extension from the optimal predecessor yields the optimal value at $D[i]$. The terminal state $D[N][n_{16}][n_8][n_4]$ is therefore the global minimum for the stated objective and quota constraints.

B. Greedy Baseline

The greedy comparator starts with every coefficient at 4 bits and repeatedly selects the available 4-to-8 or 8-to-16 upgrade with the greatest immediate reduction in squared coefficient error, subject to remaining quota. It enforces the same final quota counts as the DP.

The greedy algorithm was implemented as `adaptive_precision_error_greedy.m`. It is described in Section V alongside the 108-configuration test results.

V. Software Validation

A. Exhaustive Consistency Checks

The DP implementation was checked against exhaustive enumeration on 29 selected small cases (3–5 taps, various quotas). All costs agreed within floating-point tolerance (maximum absolute discrepancy $\sim 10^{-15}$). The greedy implementation also matched the exhaustive optima on those 29 cases.

B. 108-Configuration Test Matrix

A larger matrix covers 3 filter types (lowpass/highpass/bandpass) $\times$ 4 cutoff/center settings $\times$ 3 tap counts (8/16/32, orders 7/15/31) $\times$ 3 quotas, yielding 108 configurations. These are configurations of the same underlying approach, not 108 independent filter designs.

Across the 108 cases:

- DP and greedy both satisfy quota constraints in every case.
- The greedy matched the DP objective in 85/108 = 78.7% of configurations; median excess cost 0%, mean 37.77%, max 2166.37%.
- Maximum absolute objective gap: 2.012×10^{-6}.

These are descriptive statistics; they do not imply a general performance claim for the greedy method.

Timing was measured on MATLAB R2025a (Intel Core i5-12500H) with warmup and five repetitions. Median DP runtime was 0.047 ms; median greedy was 0.183 ms. A separate paired assignment-baseline run measured median DP 0.02615 ms and MILP 2.3322 ms. These are implementation-specific timings; they do not establish general runtime advantages.

C. Exact Assignment Baseline

All 108 configurations were solved as binary assignment MILP using MATLAB `intlinprog` (R2025a Optimization Toolbox). Every solve returned a positive exit flag, zero absolute and relative MIP gap, and used one branch-and-bound node. The maximum absolute objective discrepancy between MILP and DP across all 108 cases was 4.34×10^{-19} (floating-point evaluation). This confirms the equivalence of the DP and the exact assignment formulation but does not evidence algorithmic novelty.

VI. FPGA Case Studies

A. Eight-Tap Filter

An 8-tap direct-form FIR was implemented in Verilog for three coefficient sets: A (all 16-bit), B (all 8-bit), and C ($[4, 16, 8, 8, 8, 8, 16, 4]$). Bit-exact integer-reference simulation covered 157 cycles with zero mismatches for all three configurations.

Six synthesis/implementation runs (A1/A2, B1/B2, C1/C2; forced-DSP and auto-mapping) were completed on XC7A35T-2FGG484 with Vivado 2022.2.2 using a 10 ns clock and assumed 2 ns I/O delays. Table I reports routed resources.

TABLE I
EIGHT-TAP ROUTED FPGA RESOURCES (CORRECTED RTL)

Run	DSP	$\text{LUT}_{\text{logic}}$	LUT_{mem}	FF	Core WNS (ns)
A1/A2	8	100	0	109	5.952
B1	6	124	0	189	6.500
B2	4	166	16	279	6.725
C1/C2	6	132	0	175	6.323

Core register-to-register WNS is positive in all runs; overall WNS is negative (-0.213 to -0.695 ns) on paths ending at output ports under assumed 2 ns output delay. These are not board-level timing signoff results. FIR-only DRC reports NSTD-1 and UCIO-1 Critical Warnings on unconstrained top-level ports.

Board verification. Three bitstream configurations (A, B, C) were programmed to a legacy ACX750 development board (XC7A35T-2FGG484) via JTAG. UART-paced testing delivered the same 157-sample input sequence to each configuration; each was freshly programmed and tested twice. Across 3 configurations $\times$ 2 runs $\times$ 157 samples = 942 bit-exact comparisons against integer-precision reference values, zero mismatches were observed. This establishes functional correctness under the tested operating point; it does not measure maximum frequency, throughput, or power.

The 8-tap datapath uses signed 16-bit samples with 14 fractional bits. A w-bit signed coefficient has $w-1$ fractional bits; its $16+w$-bit product is sign-extended to 32 bits and left-shifted by $16-w$ to 29 fractional bits. The three registered adder levels use 33, 34, and 35 bits, respectively. The reference model specifies a five-cycle valid delay. The RTL performs no saturation or fractional rounding in this path; fixed-width overflow wraps. Input and coefficient quantization in the reference preparation uses rounding and clipping to representable ranges. These details concern functional validation, not measured Fmax.

B. Forty-Eight-Tap Filter

A 48-tap equiripple low-pass FIR (passband 0–0.3π, stopband 0.4π–π, $\pm$0.25 dB passband, ≤ -40 dB stopband) was designed with MATLAB `firpm`. For quota $[24, 16, 8]$, the DP allocation has coefficient SSE 1.8594×10^{-5} and a computed stopband peak of -41.30 dB (meets the -40 dB limit); a same-quota greedy allocation has SSE 6.6658×10^{-5} and -35.78 dB (fails). Two other quotas ($[16, 24, 8]$ and $[12, 24, 12]$) fail the stopband limit even under DP, showing that coefficient-SSE optimality does not guarantee spectral compliance.

Quotas count the 48 coefficient positions before folding. The allocators do not impose paired-width constraints: for quota $[24, 16, 8]$, the DP allocation has two mirror pairs with different width labels (1/48 and 4/45), and greedy has six (1/48, 2/47, 4/45, 5/44, 7/42, and 10/39). In every such pair both quantized values are zero. After fixed point alignment the quantized vectors are therefore exactly mirror symmetric (maximum mirror error zero), which is the condition used by the folded RTL. The RTL generator checks this before emitting the folded design. Each shared product uses the width and stored constant of the first member of the pair (positions 1–24); mismatched mirror labels only occur for zero values. The folded datapath uses 24 pair sums and 24 coefficient products; the reported quota remains the pre-fold count.

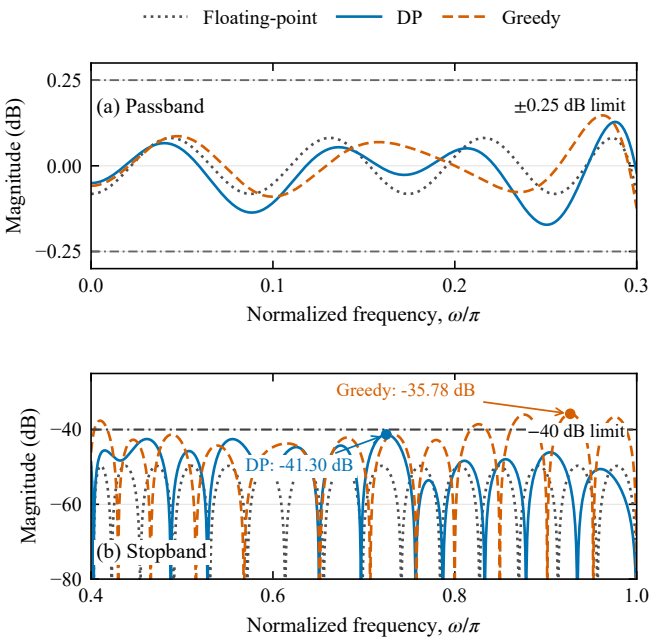


Fig. 1. Computed frequency responses of the 48-tap FIR for quota $[24, 16, 8]$: (a) passband detail and (b) stopband detail. Gray dash-dot lines mark the stated limits. DP and greedy stopband peaks are -41.30 and -35.78 dB, respectively. These curves are calculated from the saved coefficients, not measured on hardware.

Figure 1 compares the computed responses for the first quota. Both allocations meet the passband limit, but only DP meets the stopband limit in this case.

Eight RTL variants (DP, greedy, uniform 16-bit, uniform 8-bit; direct and symmetric-folded) passed XSim simulation over 2,500 cycles with 1,861 valid outputs per design (14,888 total output comparisons) against an independent 48-term integer-convolution reference. Four symmetric-folded and direct-form routes were completed on the same XC7A35T part. Table II reports routed resources for the DP and greedy symmetric-folded designs.

The tested RTL uses signed 16-bit input samples with 14 fractional bits and signed coefficient words with one sign bit and $w-1$ fractional bits. Folded products use a signed 17-bit pair sum. Products and every registered partial-sum node at tree levels 1–5 are signed 38-bit values with 29 fractional bits; the output is the signed 38-bit value at level 5, also with 29 fractional bits (9 remaining sign/integer bits). Product alignment uses exact left shifts, with no rounding or saturation; fixed-width overflow wraps. The generated vectors remain within the representable 38-bit output range. The eight registered arithmetic stages give a seven-edge accepted-input to valid-output difference in the testbench. These statements describe the tested RTL and are not a general resource model for arbitrary widths.

The response-compliant DP allocation uses more slice LUTs

TABLE II
FORTY-EIGHT-TAP ROUTED RESOURCES FOR QUOTA $[24, 16, 8]$

Design	DSP	LUT_L	LUT_M	Slice LUT	FF	Reg–reg (ns)
DP, folded	21	322	32	354	1066	4.344
Greedy, folded	21	200	16	216	997	5.189
DP, direct	38	654	80	734	1106	5.763
Greedy, direct	39	412	64	476	1043	4.879

Overall WNS (row order): -0.213, -0.430, -0.435, -0.213 ns. Negative WNS on assumed output-delay paths.

(354 vs. 216) and flip-flops (1066 vs. 997) than the non-compliant greedy allocation; this is a bounded resource–response tradeoff for one filter and one quota, not a general superiority claim. No 48-tap board test, measured Fmax, sustained throughput, or power measurement was performed.

VII. Robustness Study: Five Filter Specifications

A. Designs and Quotas

To assess whether the 48-tap base-case observation generalizes, five FIR designs were tested under three proportional quotas each (15 design/quota combinations). The designs are:

- A: 48-tap, passband 0–0.3π, stopband 0.4π–π, -40 dB (base case)
- B: 48-tap, same bands, -50 dB stopband limit
- C: 48-tap, passband 0–0.2π, stopband 0.3π–π, -40 dB (narrower transition)
- D: 36-tap, same bands and limit as A (replaced the originally planned 32-tap case, which did not meet -40 dB as a floating-point prototype)
- E: 64-tap, same bands and limit as A

All five floating-point prototypes satisfy their stated passband and stopband limits. Quotas are proportional to the 48-tap base case ($[24, 16, 8]$, $[16, 24, 8]$, $[12, 24, 12]$), scaled proportionally to the tap count and adjusted to sum exactly to that count. Some scaled quotas are odd (for example, $[22, 32, 10]$ for 64 taps).

B. Results

Table III reports the coefficient SSE and computed stopband peak for each design/quota/method combination. DP and MILP coefficient-SSE optima agree in every case.

C. Interpretation

Figure 2 shows the stopband margin for every tested combination and flags passband violations. DP meets both response limits in 2 of 15 combinations; greedy meets them in 0. The two DP passes occur for the highest-precision quota in cases A and E. This is not a universal frequency-response advantage: in case C's first quota, greedy has a slightly lower stopband peak (-35.37 vs. -34.86 dB), although both fail the -40 dB limit.

Lower coefficient SSE does not imply lower stopband peak or compliance with a response specification. These results support a bounded observation about five prototypes and three proportional quotas only; they do not establish general performance claims. All results are software frequency-response calculations; the 36- and 64-tap designs were not synthesized, routed, or tested on hardware.

TABLE III
ROBUSTNESS STUDY: COEFFICIENT SSE AND COMPUTED STOPBAND PEAK

Case	Quota	DP SSE	Greedy SSE	DP stop (dB)
A	$[24, 16, 8]$	$1.86{\cdot}10^{-5}$	$6.67{\cdot}10^{-5}$	-41.30
A	$[16, 24, 8]$	$5.61{\cdot}10^{-5}$	$1.07{\cdot}10^{-4}$	-35.32
A	$[12, 24, 12]$	$9.93{\cdot}10^{-5}$	$1.25{\cdot}10^{-4}$	-31.45
B	$[24, 16, 8]$	$4.02{\cdot}10^{-5}$	$8.42{\cdot}10^{-5}$	-35.55
B	$[16, 24, 8]$	$9.35{\cdot}10^{-5}$	$1.24{\cdot}10^{-4}$	-34.02
B	$[12, 24, 12]$	$1.56{\cdot}10^{-4}$	$1.56{\cdot}10^{-4}$	-32.42
C	$[24, 16, 8]$	$5.47{\cdot}10^{-5}$	$9.62{\cdot}10^{-5}$	-34.86
C	$[16, 24, 8]$	$8.18{\cdot}10^{-5}$	$1.16{\cdot}10^{-4}$	-35.32
C	$[12, 24, 12]$	$1.32{\cdot}10^{-4}$	$1.51{\cdot}10^{-4}$	-31.85
D	$[18, 12, 6]$	$2.22{\cdot}10^{-5}$	$2.22{\cdot}10^{-5}$	-35.60
D	$[12, 18, 6]$	$5.94{\cdot}10^{-5}$	$5.94{\cdot}10^{-5}$	-30.75
D	$[10, 16, 10]$	$3.02{\cdot}10^{-4}$	$3.02{\cdot}10^{-4}$	-28.16
E	$[32, 22, 10]$	$2.20{\cdot}10^{-5}$	$7.67{\cdot}10^{-5}$	-40.76
E	$[22, 32, 10]$	$5.50{\cdot}10^{-5}$	$8.08{\cdot}10^{-5}$	-36.78
E	$[16, 32, 16]$	$8.68{\cdot}10^{-5}$	$9.68{\cdot}10^{-5}$	-32.85

VIII. Response-Aware Allocation: Feasibility Investigation

This section documents an investigation into whether the coefficient-SSE objective could be replaced by a frequency-response-aware objective, and whether the resulting allocation could be solved exactly under exact quota constraints.

A. Motivation and Mathematical Structure

The coefficient-SSE objective $\sum_i e_i^2$ is (by Parseval's theorem) proportional to the unweighted integrated squared frequency-response error over the full band. To focus on stopband compliance, the natural objective is the stopband-weighted integrated squared complex-response error $\mathbf{e}^\top R\mathbf{e}$, where

$$R_{ij} = \frac{1}{\pi} \int_{\Omega_{\text{stop}}} \cos(\omega(i-j))\, d\omega \tag{8}$$

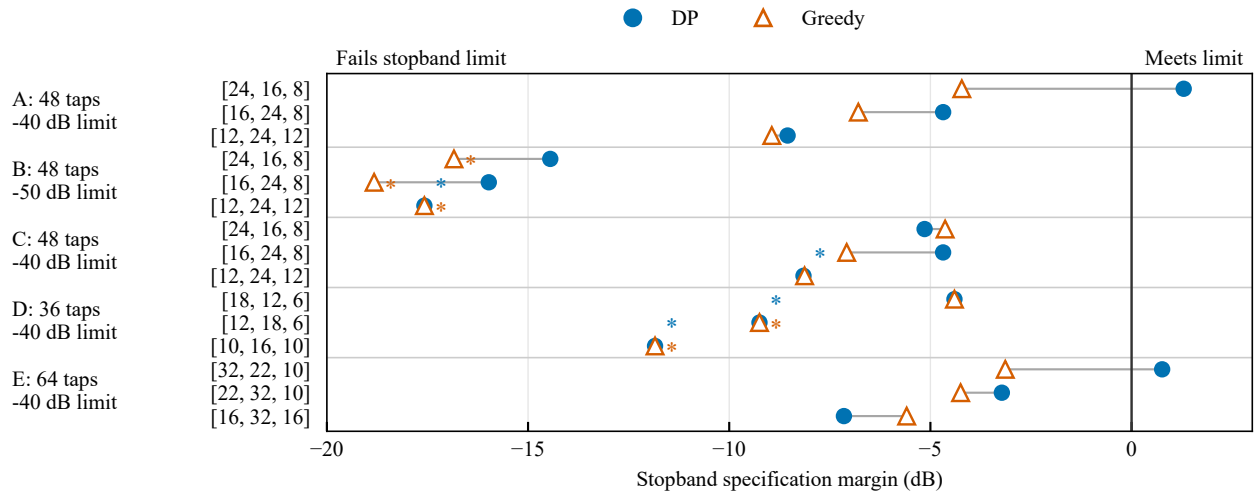
and Ω_{stop} is the stopband interval.

Three key properties were established:

1) R_{ii} is the same for all tap positions i: the diagonal weight equals the stopband bandwidth fraction and does not vary with tap index. Diagonal approximation therefore produces a uniform scaling of the SSE objective and does not change the allocation.
2) R has nonzero off-diagonal elements (magnitudes 6×10^{-6} to 0.30 for the 48-tap case), so the objective $\mathbf{e}^\top R\mathbf{e}$ is not separable in the tap errors e_i. The existing quota DP, which assumes a separable objective, cannot directly solve the weighted formulation.
3) Fang et al. [11] also studied FIR allocation with a frequency-domain objective (minimax), so the objective type alone does not establish novelty.

B. Small-Scale Exhaustive Screening

To assess whether handling the cross-terms changes the optimal allocation, exhaustive search was run on five 8-tap filters under two quotas each (10 cases). Each case enumerated all feasible assignments (420–560 per case), found the SSE-minimizing allocation and the $\mathbf{e}^\top R\mathbf{e}$-minimizing allocation, and compared them.



Quota counts are [16-bit, 8-bit, 4-bit]. * Passband deviation > 0.25 dB.

Fig. 2. Computed stopband margin for all 15 design/quota combinations (limit minus peak; nonnegative means pass). Case B uses −50 dB; others use −40 dB. Asterisks mark passband deviations above 0.25 dB. DP meets both limits in 2/15 cases and greedy in 0/15. Responses are computed.

In 6 of 10 cases, the two optima differed and the weighted-optimal allocation had a lower stopband peak. However, the absolute stopband peaks for these 8-tap filters are poor (−5 to −20 dB), and the improvements are small (0.2–1.2 dB). The 10 cases arise from 5 filter designs × 2 quotas; independence is limited.

C. Forty-Eight-Tap MIQP Attempt

To test whether the weighted objective can be solved exactly at realistic filter scales, the objective was linearized using binary product variables. For each pair $(i, k), (j, l)$ with $i < j$, a product variable $y = x_{i,k}x_{j,l}$ was introduced with standard McCormick inequalities. The resulting MILP had:

- 144 width-selection binary variables
- 10,152 product binary variables
- 51 equality constraints
- 30,456 McCormick inequality constraints

MATLAB `intlinprog` was warm-started with the SSE-DP solution. After approximately 93 seconds, the root-node LP lower bound was approximately -5.637×10^{-4} against an incumbent of 7.4603×10^{-6} (gap ≈ 7,656%), with no branch-and-bound nodes explored. The run was manually interrupted after approximately 93 seconds under the pre-declared slow-progress stopping condition. The root bound changed negligibly from approximately -5.6371×10^{-4}. Ctrl+C terminated MATLAB before it returned an exit flag or final allocation. The negative LP bound reflects weak LP relaxation from the McCormick linearization; it is not a property of the objective.

D. Summary and Decision

Under the pre-declared stopping conditions (stop if progress is too slow; otherwise cap the run at 1 hour), this branch was closed:

> *STOP: The direct pairwise-product MILP has a weak LP relaxation that did not certify global optimality in the observed run. No response-aware allocation or improvement claim follows from the attempted runs.*

This outcome does not imply that response-aware allocation with exact quota constraints is mathematically intractable. A dedicated MIQP solver (e.g., Gurobi or CPLEX) or a stronger relaxation strategy may certify global optimality at 48-tap scale. This remains future work conditioned on solver availability.

IX. Conclusion

This report evaluated exact-quota mixed-precision FIR quantization as an instance of unit-capacity transportation. The DP implements a known cost-constrained optimization principle; no algorithmic novelty is claimed. Its coefficient-SSE optimum was verified by 29 exhaustive checks and 108 MILP comparisons (maximum discrepancy 4.34×10^{-19}).

Hardware evidence comprises six routed 8-tap runs and 942 bit-exact board comparisons, plus four routed 48-tap runs and 14,888 RTL output comparisons. In the 48-tap base case, DP meets the stopband limit where same-quota greedy fails, at the cost of more LUTs and registers. Across five designs and 15 design/quota combinations, DP meets both response limits in two cases and greedy in none; greedy nevertheless has a lower stopband peak in some failed cases. Coefficient-SSE optimality therefore does not guarantee spectral compliance.

The band-weighted response objective contains cross-terms, while its uniform diagonal approximation leaves the SSE allocation unchanged. The 48-tap binary-product MILP had a weak root relaxation and was manually stopped after about 93 seconds under the pre-declared slow-progress condition. No solver exit flag, final allocation, or optimality certificate was returned. This observed run does not establish mathematical intractability.

Future work includes tighter response-aware formulations or dedicated MIQP solvers, and 48-tap board testing, power measurement, and Fmax characterization. The released code

and data preserve both the positive validation results and these limitations.

Code and Data Availability

The source code, experimental data, FPGA RTL, and test vectors are available at https://github.com/Bairuirui68/exact-quota-mixed-precision-fir.

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